

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer – 2025

Course: B. Tech Branch : Mechanical Engineering **Semester :IV**

Subject Code & Name: BTMEC403_Y19 **Strength of Materials**

Max Marks: 60

Date : 07/07/2025

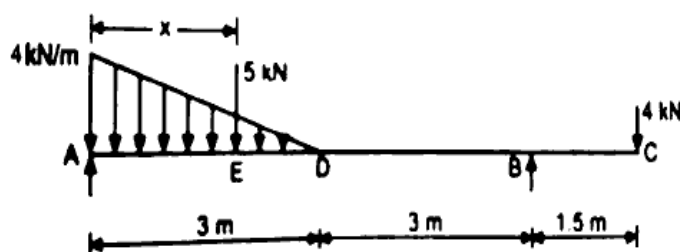
Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Which law is also called as the elasticity law? a. Poisson's law b. Bernoulli's c. Stress law d. Hooke's law	CO 1	1
2 Where in the stress-strain curve, the Hooke's law is valid? a. Elastic range b. Valid everywhere c. Necking region d. Strain hardening region	CO 1	1
3 The property by which a body returns to its original shape after removal of the force is called _____ a. a) Plasticity b. Elasticity c. Ductility d. Ductility	CO2	1
4 The materials which have the same elastic properties in all directions are called.. a. Isotropic b. Brittle c. Homogeneous d. Hard	CO2	1
5 Which stress comes when there is an eccentric load applied? a. Shear stress b. Bending stress c. Tensile stress d. Thermal stress	CO1	1
6 What is the expression of the bending equation? a. $M/I = \sigma/y = E/R$ b. $M/R = \sigma/y = E/I$ c. $M/y = \sigma/R = E/I$ d. $M/I = \sigma/R = E/y$	CO2	1
7 What is the formula for shear stress? a. Shear resistance / shear area b. Force/unit area c. Tensile stress d. Thermal stress	CO1	1
8 Which stress comes when there is an eccentric load applied? a. Bending stress b. Shear stress c. kg d. kg/cm ²	CO1	1
9 The ultimate tensile stress of mild steel compared to ultimate compressive stress is a. same b. more c. less d. unpredictable	CO2	1
10 The neutral axis is of the cross-section a % beam is that axis at which the bending stress is a. zero b. minimum c. maximum d. infinity	CO2	1

- 11 Euler's formula holds good for
a. short columns b. long columns c. both short & long columns d. weak columns CO1 1
- 12 . Which type of stress is primarily concerned with twisting forces?
a. Tensile stress b. Shear stress c. Compressive stress d. Bending stress CO1 1
- Q. 2 Solve the following.** 12
- A) Define the following terms:
i) Poisson's Ratio 2) Modulus of Elasticity 3) Hook's Law CO1 6
- B) A cylinder 150 mm in diameter, 300 mm in length is subjected to axial compressive load of 180 kn which cause increase in diameter 0.0952 mm and a decrease in length by 0.64 mm. Calculate poison's ratio and modulus of elasticity. CO3 6
- Q.3 Solve the following.** 12
- A) Explain middle third rule and kernel of a section. CO1 6
- B) Water under pressure of 10 Mpa is suddenly admitted on to a plunger of 10 mm diameter, attached to a rod 228 mm diameter, 3 m long. Find the maximum instantaneous stress and deformation of the rod if $E = 210 \text{ GPa}$. CO2 6
- Q. 4 Solve Any Two of the following.** 12
- A) Derive the flexural formula. CO2 6
- B) A square column of 100 mm x 100 mm cross section has a concentric longitudinal hole of 50 mm diameter. The length of the column is 5 m. one end fixed and other hinged. Determine the Euler's buckling load assuming $E = 200 \text{ Gpa}$. CO3 6
- C) Find Power transmitted by a shaft having 60 mm diameter rotating at 150 r.p.m, if maximum permissible shear stress = 80 Mpa CO3 6
- Q.5 Solve Any Two of the following.** 12
- A) Derive the equation of torsion for circular shaft. CO2 6
- B) A simply supported beam ABC is loaded as shown in figure. Determine the location at which concentrated load 5 kN must act from end A to make the reactions at A and B equal. Draw SFD and BMD. CO4 6



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| C) A Steel shaft of solid circular cross section has to transmit 200 kW at 180 r.p.m. The maximum shear stress is not to exceed 50 MPa and the angle of twist must not be more than 1.60 in a length of 2.6 m. Design suitable diameter of shaft. Take $G=80$ GPa. | CO3 | 6 |
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Q. 6 Solve Any Two of the following.

12

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| A) Derive the equation of shear force, bending moment and intensity of loading. | CO4 | 6 |
| B) Derive the shear stress formula. | CO4 | 6 |
| C) Derive the relation between $\tau_{\max}$ and τ_{avg} for | | 6 |
| i) Rectangular section | CO3 | |
| ii) Circular section | | |

***** End *****